MarketPulse: Leakage-Controlled Market-Regime Classification and Regime-Change Probability Estimation with an HMM-Augmented, Calibrated Ensemble

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Abstract

We study next-day classification of six market regimes (Bullish/Sideways/Bearish $\times$ Low/High volatility) and the probability that the regime label changes within 5, 10 or 20 trading days, using 7,302 real daily observations of the S&P 500, VIX and Brent crude (1990–2018). The pipeline enforces chronological train/validation/test splits with an embargo, causal features, forward-filtered (not smoothed) hidden Markov model (HMM) states, walk-forward hyperparameter search with Optuna, and validation-only calibration. A temperature-calibrated soft-voting ensemble over HMM-augmented features reaches a test log loss of 0.391 (95% block-bootstrap CI 0.335–0.453), versus 0.783 for the persistence baseline, while persistence retains the best accuracy (0.861 vs. 0.843) and macro-F1 (0.837 vs. 0.800). Ablations show that the HMM state probabilities, 12 additional engineered features and cross-asset (VIX/Brent) covariates give no measurable test improvement, because the labels are deterministic functions of 60-day momentum and volatility (SHAP confirms these two features dominate). Error analysis shows 88% of errors involve the trend dimension and error rates of 36% for days closest to a label threshold versus 2.5% for the farthest. The model is served by a FastAPI application in a Docker container. All code, tables and figures are reproducible from the repository.

Index Terms— market regimes, hidden Markov model, gradient boosting, probability calibration, Optuna, walk-forward validation, error analysis

I Introduction

Regime awareness—knowing whether markets are trending or range-bound, calm or turbulent—is central to risk management and asset allocation [1, 3, 4]. Two practical problems recur: (i) *classify* the current/next regime, and (ii) *estimate how likely a regime change is* over a short horizon. Published studies often evaluate on random splits, smooth HMM states with future data, or tune on the test period, all of which inflate results. This work builds a deliberately conservative pipeline and reports honest comparisons, including where simple baselines win.

Contributions. (1) A leakage-controlled pipeline (causal features, embargoed chronological splits, filtered HMM, validation-only calibration) with unit tests that perturb the future. (2) A feature-engineering and dimensionality-reduction study (33 causal features in 7 groups, PCA, t-SNE). (3) Optuna walk-forward tuning with a validation gate and experiment logging. (4) Benchmarks against re-implemented literature baselines with block-bootstrap confidence intervals, paired tests and a structured error analysis. (5) A containerised FastAPI inference service.

II Related Work and Baselines

Hamilton's Markov-switching model [1] and HMM inference [2] underpin regime-switching asset-allocation work [3, 4]. Volatility regimes are classically forecast with RiskMetrics EWMA [5] and GARCH [6]; trend regimes with moving-average rules [7]. Gradient-boosted trees [8, 9], tuned by Bayesian optimisation [10], are strong tabular learners, and temperature scaling [11] is a simple post-hoc calibrator. SHAP [12] provides attributions. We re-implement these *method families* on a common task and split; we do not reproduce any paper's exact experiments and quote none of their numbers, as datasets, labels and horizons differ.

III Data

III.1 Dataset and Provenance

The bundled dataset contains 7,302 trading days (2 Jan 1990–21 Dec 2018) of S&P 500 OHLCV, the CBOE VIX close and Brent spot (USD/bbl). Sources are listed in the repository data card; the S&P 500 series ends in Dec. 2018, so the test period excludes the 2020 and 2022 episodes. Cleaning removes duplicates, non-positive prices and $high < low$ rows (none occurred), forward-fills cross-asset gaps of at most five rows, and leaves 78 early Brent gaps missing (excluded by feature warm-up rather than imputed). Figure 1 shows the series with regimes and VIX.

III.2 Regime Labels and Targets

Trend is Bullish/Sideways/Bearish from the trailing 60-day return using the 30th/70th percentiles of the *training* period (thresholds -0.13% and $+5.38\%$); volatility is High if 60-day annualised realised volatility exceeds the training median (12.5%). This yields six classes. The supervised target is the label at $t{+}1$ given features at t. The transition target is $\mathbb{1}[\text{label changes within } h \text{ days}]$, $h \in \{5, 10, 20\}$, with 43%, 60% and 78% prevalence on test.

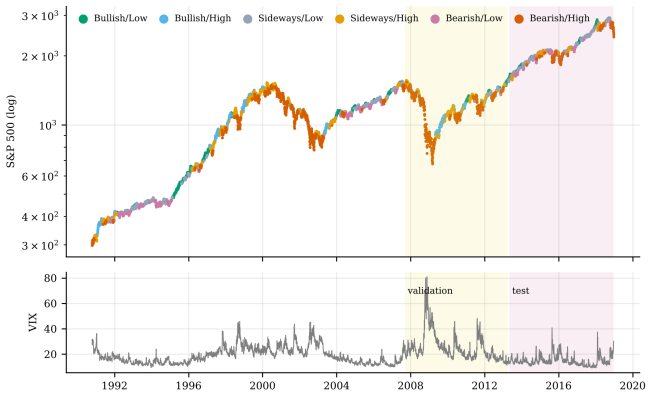


Figure 1: S&P 500 coloured by rule-based regime, VIX below, shaded validation/test windows.

III.3 Splits

Chronological 60/20/20 splits with a 20-row embargo: train 1990-10-15 to 2007-08-10 (4,241 rows), validation 2007-09-11 to 2013-04-04 (1,401), test 2013-05-03 to 2018-12-20 (1,420). Label thresholds, scaler, PCA, HMM, calibrators and tuning use training (or validation where stated) rows only.

IV Feature Engineering and Dimensionality Reduction

IV.1 Features

Thirty-three causal features in seven groups: returns (5), volatility windows (4), momentum/RSI/drawdown (5), log-volume z-score (1), *advanced volatility* (6: Parkinson and Garman–Klass range estimators, EWMA σ, vol-of-vol, skewness, kurtosis), *trend* (6: SMA gaps, MACD histogram, Bollinger %B, return autocorrelation, up-day ratio) and *cross-asset* (6: VIX level, 5-day change, 60-day z-score, implied-minus-realised spread; Brent 20-day return and volatility, lagged one day for publication delay). A unit test corrupts all columns after row 3000 and asserts earlier features are unchanged. Figure 2 shows strong block structure (volatility estimators correlate > 0.9).

IV.2 PCA and t-SNE

PCA on standardised, clipped training features needs 18 components for 95% variance; PC1 (29.2%) and PC2 (18.8%) load on volatility estimators and short-term momentum, respectively (Fig. 3). Regimes overlap heavily (silhouette -0.03), consistent with labels being threshold bands of continuous signals rather than natural clusters; t-SNE (Fig. 4) shows the same continuum with a distinct high-volatility arm. As model input, PCA hurts: logistic regression on 18 PCs scores macro-F1 0.697 versus 0.769 on raw features, and 3 PCs only 0.419 (Table 1). Dimensionality reduction is therefore used for diagnosis, not prediction.

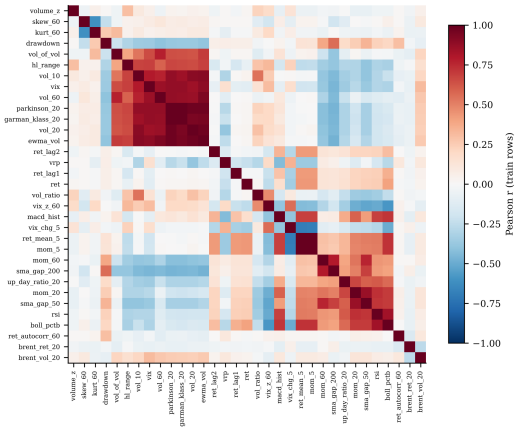


Figure 2: Clustered feature correlation (training rows).

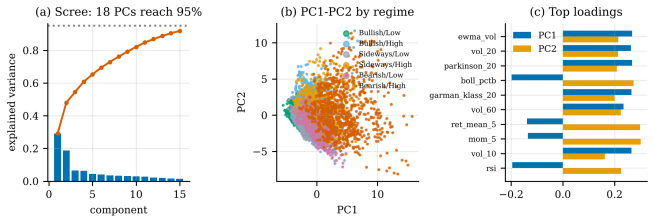


Figure 3: PCA: scree, PC1–PC2 by regime, loadings.

V Methods

V.1 Filtered HMM

A 3-state Gaussian HMM (full covariance, best of four EM restarts) is fitted on training rows over [return, 20-day volatility, 20-day momentum]. State probabilities are computed with the forward algorithm, $P(s_t \mid x_{1:t})$, plus the one-step predictive $P(s_{t+1} \mid x_{1:t})$; hmmlearn's smoothed posteriors are deliberately avoided because they use future observations. Test: truncating the series leaves earlier filtered probabilities identical (tolerance 10^{-9}). State names (Bullish/Sideways/Bearish $\times$ volatility) are assigned from training-period state means by rank.

V.2 Models

Logistic regression, random forest, XGBoost and LightGBM are wrapped so probabilities always span all six classes. The *soft-voting ensemble* averages member probabilities; the *HMM-augmented ensemble* adds the six HMM columns; the *calibrated hybrid* applies temperature scaling ($T = 0.838$, fitted on validation log loss) [11]. Transition models are LightGBM classifiers per horizon on features plus current regime and days-in-regime; calibration is chosen on validation among none/Platt/isotonic by forward-chained Brier score.

V.3 Hyper-parameter Optimisation

Optuna (TPE sampler, median pruner) [10] runs 25 trials per algorithm on training rows with 4 expanding-window

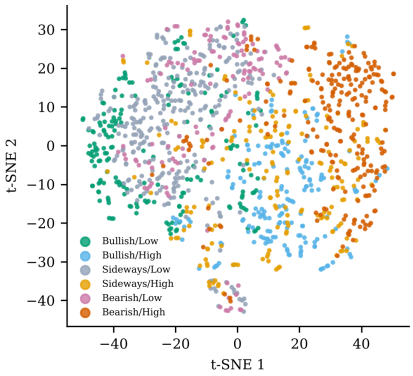


Figure 4: t-SNE of training features (1,200-point subsample).

Table 1: PCA input study (test, 6-class)

Input	#	Model	Macro-F1	Log loss
PCA-3	3	LR	0.419	1.056
PCA-8	8	LR	0.602	0.722
PCA-18 (95%)	18	LR	0.697	0.625
PCA-18 (95%)	18	LightGBM	0.568	0.885
Raw features	33	LR	0.769	0.438
Raw features	33	LightGBM	0.782	0.543

folds separated by a 20-row gap, maximising negative log loss. Tuned parameters replace defaults only if validation log loss improves (a gate against over-fitting the search). Every study, parameter set and metric is appended to `experiments/runs.jsonl` with data fingerprint, library versions and seed.

V.4 Statistical Protocol

All headline numbers are on the untouched test split. Confidence intervals use a moving-block bootstrap (20-day blocks, 300 replicates); comparisons use paired block bootstrap of metric differences on identical resamples (500 replicates).

VI Results

VI.1 Hyper-parameter Tuning

Optuna improved three of four algorithms on validation log loss (Fig. 5): random forest 0.563→0.369, XGBoost 0.383→0.353, LightGBM 0.479→0.367; logistic regression (0.371→0.388) was rejected by the gate. For the hybrid, test log loss improved from 0.440 to 0.391 and macro-F1 from 0.783 to 0.800, with the 11–12 of 25 pruned trials per boosting study saving compute.

VI.2 Six-class Benchmark

Table 2 and Fig. 6 compare 12 methods. Three findings: (i) persistence has the best accuracy and macro-F1 because regimes last for weeks, but the worst-calibrated confidence

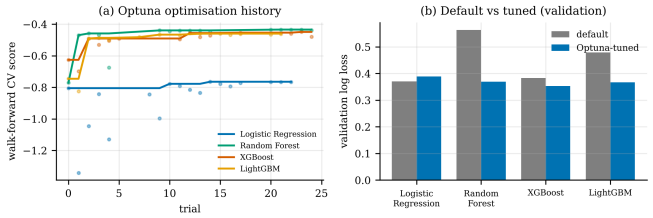


Figure 5: Optuna history (best-so-far line) and default vs. tuned validation log loss.

Table 2: Six-class next-day regime task on the test split (n = 1,420). 95% block-bootstrap CIs in brackets.

Method	Accuracy	Macro-F1	Log loss	Brier	ECE
Persistence baseline	**0.861**	**0.837** [0.790, 0.867]	0.783 [0.655, 0.939]	0.271	0.119
Soft-voting ensemble (tuned)	0.847	0.808 [0.754, 0.837]	**0.391** [0.340, 0.446]	0.216	0.031
Calibrated hybrid (ours)	0.843	0.800 [0.750, 0.823]	**0.391** [0.335, 0.453]	0.220	**0.020**
XGBoost (tuned)	0.849	0.815 [0.759, 0.846]	0.395 [0.342, 0.452]	**0.214**	0.021
HMM-augmented ensemble	0.843	0.800 [0.750, 0.823]	0.399 [0.348, 0.457]	0.220	0.031
Random forest (tuned)	0.856	0.825 [0.771, 0.858]	0.410 [0.358, 0.465]	0.216	0.038
LightGBM (tuned)	0.837	0.798 [0.743, 0.827]	0.416 [0.354, 0.484]	0.231	0.034
Logistic regression	0.820	0.769 [0.717, 0.802]	0.438 [0.378, 0.507]	0.254	0.016
MLP (64-32, L_2 tuned on val.)	0.813	0.767 [0.707, 0.806]	0.443 [0.378, 0.521]	0.256	0.024
PCA (18) + logistic regression	0.760	0.697 [0.629, 0.736]	0.625 [0.520, 0.733]	0.347	0.022
HMM-based (state→next label)	0.477	0.268 [0.218, 0.307]	1.356 [1.221, 1.526]	0.660	0.111
Majority class	0.396	0.095 [0.081, 0.110]	1.691 [1.652, 1.726]	0.797	0.160

(ECE 0.119, log loss 0.783); (ii) tuned tree models, the ensemble and the hybrid are statistically indistinguishable from one another on log loss (hybrid vs. XGBoost $\Delta = -0.004$, 95% CI $[-0.016, 0.012]$) yet significantly better than persistence ($\Delta = -0.395$, CI $[-0.496, -0.308]$), logistic regression (-0.046), LightGBM (-0.026) and MLP (-0.052); (iii) the standalone HMM-based classifier (macro-F1 0.268) cannot resolve six classes from three states, and PCA+LR loses 0.10 macro-F1. The hybrid has lower ECE (0.020) than the uncalibrated ensemble (0.031); temperature scaling gives a small but significant log-loss gain over the HMM-augmented ensemble ($\Delta = -0.008$, $p = 0.024$).

VI.3 Literature-method Baselines on Sub-tasks

Volatility regime (binary, Table 3): the hybrid collapsed to $P(\text{High})$ matches the 60-day realised-volatility score (log loss 0.053 each; ROC-AUC 0.999) and clearly beats GARCH(1,1)-t (0.257), RiskMetrics EWMA (0.279) and VIX level (0.576) used as single predictors with Platt scaling fitted on train+validation. This is expected: the label *is* thresholded 60-day volatility, so the 20-day-horizon forecasters carry less of the signal. **Trend regime** (3-class): persistence (macro-F1 0.864) > ensemble (0.853) > hybrid (0.851) ≫ SMA-50/200 rule (0.296). The rule scores poorly because it is not designed to reproduce a quantile-defined label; it is a reference point, not a fair rival.

VI.4 Regime-change Probabilities

On the harder transition task (Table 4) LightGBM beats the constant, days-in-regime hazard, logistic regression and HMM-implied baselines in PR-AUC at 5 and 10 days; at 5 days PR-AUC is 0.636 [0.550, 0.728] versus 0.427 prevalence,

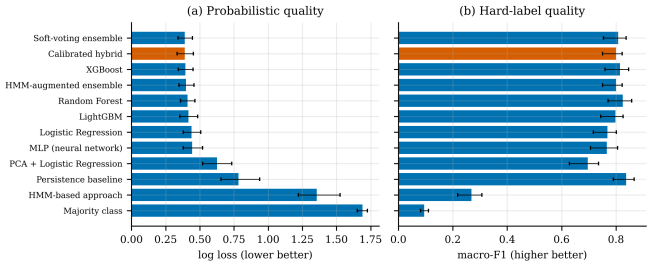


Figure 6: Test log loss and macro-F1 with 95% block-bootstrap CIs; hybrid in orange.

Table 3: Sub-task benchmarks (test)

Volatility regime (binary) Method	Acc.	AUC	Log loss
Hybrid (collapsed)	0.983	0.999	**0.053**
Realised vol 60d	0.988	0.998	**0.053**
Persistence	0.988	0.987	0.067
GARCH(1,1)-t	0.905	0.956	0.257
EWMA σ	0.895	0.948	0.279
VIX level	0.752	0.834	0.576
Trend regime (3-class) Method	Acc.	Macro-F1	
Persistence	0.870	0.864	
Hybrid (collapsed)	0.859	0.851	
HMM-based	0.539	0.479	
SMA-50/200 rule	0.287	0.296	

but intervals overlap with the empirical hazard (0.591), so the evidence of superiority is weak at this sample size. Validation-selected calibration chose "none" for all horizons; the post-hoc study agrees: Platt and isotonic calibrators fitted on the 2007–2013 validation window *worsen* test Brier (e.g. 10-day: 0.223 none, 0.228 Platt, 0.235 isotonic) because the validation window had different change frequencies. Calibrating on a window unlike deployment can hurt.

VI.5 Ablations

Feature groups (single LightGBM, test macro-F1, Fig. 7): returns only 0.269; +volatility 0.409; +momentum 0.783; +volume 0.787; +advanced volatility 0.778; +trend 0.776; +cross-asset 0.782. Momentum is decisive; later groups add nothing within noise. For the transition task the extra groups were slightly *harmful* (5-day PR-AUC 0.671 core, 0.652 with advanced, 0.637 full; 3 seeds, sd $\leq$ 0.008). *HMM information*: ensemble without HMM columns: log loss 0.391 vs. 0.399 with them (test). SHAP (Fig. 8) explains why: `mom_60` and `vol_60`, which define the labels, carry nearly all attribution (mean |SHAP| 1.26 and 1.22 versus $\leq$ 0.13 for any other feature). We keep the extended features and HMM in the shipped model as documented, evidence-backed non-contributors rather than tuning the feature set on the test split.

Table 4: Regime change within h days (test; PR-AUC [95% CI])

Method	h=5	h=10	h=20
Prevalence	0.427	0.595	0.776
Days-in-regime hazard	0.591	0.728	0.846
HMM-implied	0.476	0.636	0.802
Logistic regression	0.570	0.748	**0.895**
LightGBM (ours)	**0.636**	**0.767**	0.886

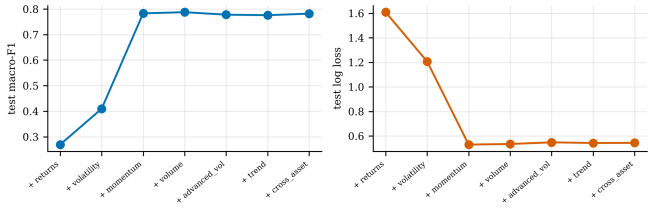


Figure 7: Cumulative feature-group ablation (test).

VI.6 Calibration

Fig. 9 shows top-class reliability: the hybrid's confidence tracks accuracy within about 0.05–0.10 for bins above 0.6 (error 4.3% at confidence $>$ 0.9 with 805 days, 52% for confidence 0.5–0.6) while persistence's implicit 98% confidence is right only 86% of the time.

VII Error Analysis

Analysis is on the 1,420 test days (hybrid error 15.7%, persistence 13.9%).

- **Which dimension:** 88% of errors are trend-only, 10% volatility-only, 2% both. Volatility regimes are nearly deterministic; the trend band is where the model errs.
- **Threshold proximity:** error rate is 36.3% for the quartile of days closest to a trend/volatility threshold and 2.5% for the farthest quartile (Fig. 10a). Most errors are unavoidable coin-flips about which side of an arbitrary cut-off tomorrow's 60-day return lands on.
- **Change days:** of 197 true regime changes, the hybrid flags 80 and catches 28 (recall 14%, precision 35%); on non-change days it is 95.7% accurate, versus 100%/0% for persistence. The model's net advantage is probabilistic quality, not change detection.
- **Confusions:** Sideways/Low→Bullish/Low (47 days, 21% of errors) and the reverse dominate; Bullish/High has the lowest recall (0.50, $n = 86$), and Bullish/High → Sideways/High is the main leak.
- **Environment and time:** errors are highest in mid-VIX regimes (20.0%) and lowest at VIX $>$ 20.6 (7.9%); by year the hybrid is worse than persistence in 2013–2016 (peak 22.2% in 2016, the Aug-2015/Jan-2016 sell-offs) and better in 2018 (13.1% vs 15.5%) (Fig. 10c).

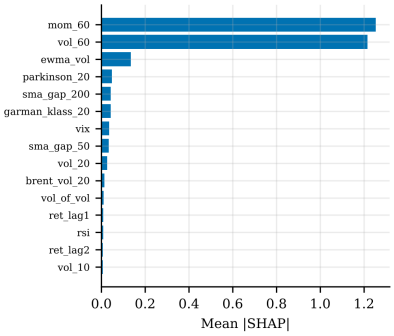


Figure 8: Mean |SHAP|, tuned XGBoost, 400 test rows.

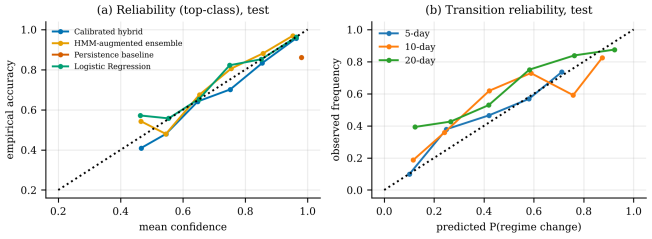


Figure 9: Reliability of regime probabilities and transition probabilities (test).

- **Overconfident errors:** the ten most confident mistakes (0.95–0.98) are mostly Bearish/Low days predicted as Sideways/Low in 2013–2018, e.g. 2014-09-24 and 2016-06-23 (the Brexit vote): abrupt moves that push the 60-day return across the bearish cut-off.

VIII System and Deployment

Application. A FastAPI service (`api/main.py`) loads a 5 MB model bundle (calibrated ensemble, HMM, label thresholds, transition models and calibrators) and exposes `GET /health`, `/model`, `/demo`, `POST /predict` (validated OHLCV+VIX+Brent history, $\geq$320 rows) and a browser demo page. Responses give the current regime and its age, next-day class probabilities, a low-confidence flag, regime-change probabilities at 5/10/20 days, and flag whether the as-of date lies in the training or validation window. Tests verify that API outputs equal the training pipeline's outputs to 2×10^{-4} and that malformed input (short history, $high < low$, missing VIX, unsorted dates) returns HTTP 422. A Streamlit dashboard (nine pages, including Feature Lab and Benchmarks & Errors) supports exploration. **Containerisation.** A `python:3.13-slim` Dockerfile installs pinned runtime dependencies, runs as non-root (uid 1000), listens on `$PORT` (7860, Hugging Face Spaces), and defines a health check; `docker-compose.yml` adds the dashboard as an optional service. Deployment steps for Hugging Face Spaces (Docker SDK) and AWS App Runner are in `docs/DEPLOYMENT.md`. Live URL: [insert after deployment]. **Reproducibility.** `python train.py --tune` re-

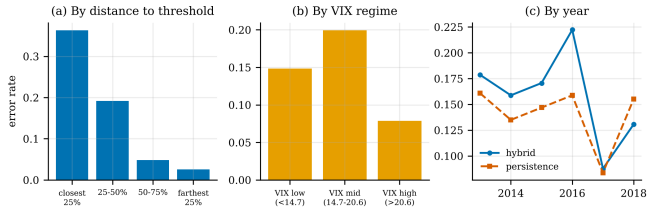


Figure 10: Error rate by distance to threshold, VIX regime and year.

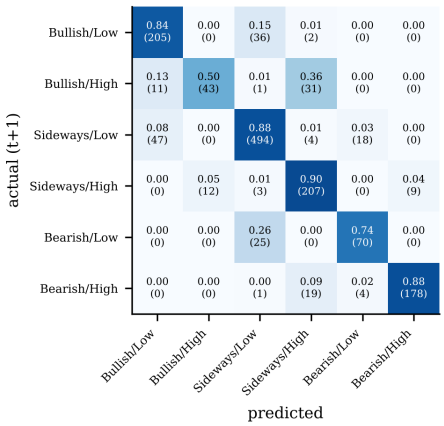


Figure 11: Row-normalised confusion matrix of the hybrid (counts in brackets).

generates every table, figure and the bundle (about 17 minutes on 2 CPU cores); 30 automated tests cover causality, splits, calibration, tuning, tracking, benchmarks, errors and the API.

IX Limitations and Ethics

(i) Labels are rule-based, so supervised accuracy largely measures persistence and threshold proximity; the task is not tradable alpha. (ii) One instrument and one test window (2013–2018, a low-volatility bull market with few crises); results may not transfer to 2020 or 2022. (iii) Bootstrap CIs account for serial dependence but not label-threshold or model-selection uncertainty. (iv) Baselines are method-family re-implementations with modest tuning, not paper replications. (v) The S&P 500 file's redistribution licence is unstated. (vi) The system is an educational tool; it gives no investment advice and no trading signal.

X Conclusion

A carefully leakage-controlled pipeline shows that calibrated tree ensembles substantially improve the probabilistic quality of regime forecasts over persistence (log loss 0.391 vs. 0.783) while persistence remains hard to beat on hard-label accuracy. HMM states, advanced features and cross-asset covariates did not help on these labels—a negative result we report instead

of tuning around. Future work: market-defined (not threshold-defined) regime targets, additional instruments and post-2018 data, and economic-value evaluation with transaction costs.

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